Product overview

Product description and product pictures

ThinkSystem SR650 V3 product overview

The SR650 V3 server (machine type 7D75 and 7D76) is a 2U two-socket (2U2S) rack server that features 4th Generation Intel Xeon Scalable processors (Intel code name: Sapphire Rapids) and is the successor to the SR650 V2 server. The chassis supports either 3.5-inch or 2.5-inch hard drives, and an expander can be used to support rear drives (2.5-inch or 3.5-inch). Single- and double-width GPUs are also supported with some limitations.

SR650 V3 specifications

Scroll down for more information

Attribute	Specifications
Form factor	2U2S rack mount
Processor	One or two 4th Generation Intel Xeon Scalable processors (codename: Sapphire Rapids) – Bronze, Silver, Gold, or Platinum level, up to 350 W (without HBM) 8xxx SKU CPUs support 4 UPIs – others support 3 UPIs
Memory	Up to 32 DIMM slots (16 DIMMs per processor) • Eight channels per CPU • Two DIMMs per channel • 32 DDR5 DIMMs – 1DPC 4800 MHz , 2DPC 4400 MHz
Disk drive bays	Front Bay: • Eight or 12 3.5-inch SATA/SAS • 12 3.5-inch AnyBay • Eight, 16, or 24 2.5-inch SATA/SAS • Eight, 16, or 24 2.5-inch NVMe • Eight or 16 2.5-inch AnyBay • 12 3.5-inch SAS/SATA with expander (supports four rear 3.5-inch)

Note: For the latest specifications, refer to the Lenovo Press product guide.

SR650 V3 specifications

Scroll down for more information

Attribute	Specifications
	• 24 2.5-inch SAS/SATA with expander (supports four rear 2.5-inch) Rear Bay: • Two or four 3.5-inch SAS/SATA / four or eight 2.5 SAS/SATA / four 2.5 Anybay • Two 7 mm Middle Bay: • Four 3.5-inch or eight 2.5-inch SAS/SATA or NVMe
Network interface	• Dedicated OCP 3.0 SFF slot with a PCIe 4.0 x16 host interface • Supports a variety of two-port and four-port adapters with 1 GbE, 10 GbE, and 25 GbE network connectivity
PCIe expansion slots	• Supports up to 10 PCIe slots (LP risers 3 and 4 have four x8 slots) • One OCP NIC 3.0
GPU support	• Three double width GPUs with limitation (max 350 W H100) • Six single width GPUs (150 W) • Eight single width GPUs w/ 75 W
Cooling	• Six fan solution • Two fan types for different system configurations: standard and high-performance

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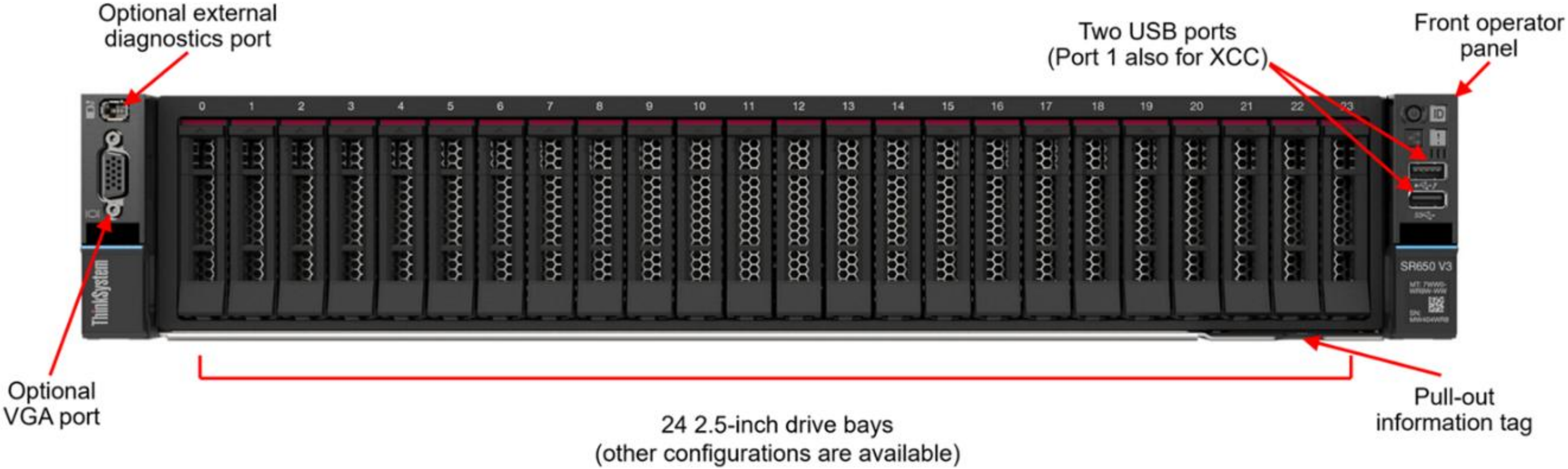
Power supplies	Up to two hot swap redundant power supplies
Storage controller	• Intel® Virtual RAID on CPU (VROC) • RAID Controllers with eight, 16, or 32 ports • Non-RAID HBA with eight or 16 ports • Internal 8i or 16i HBA/RAID • Internal expander (max 40 port output)

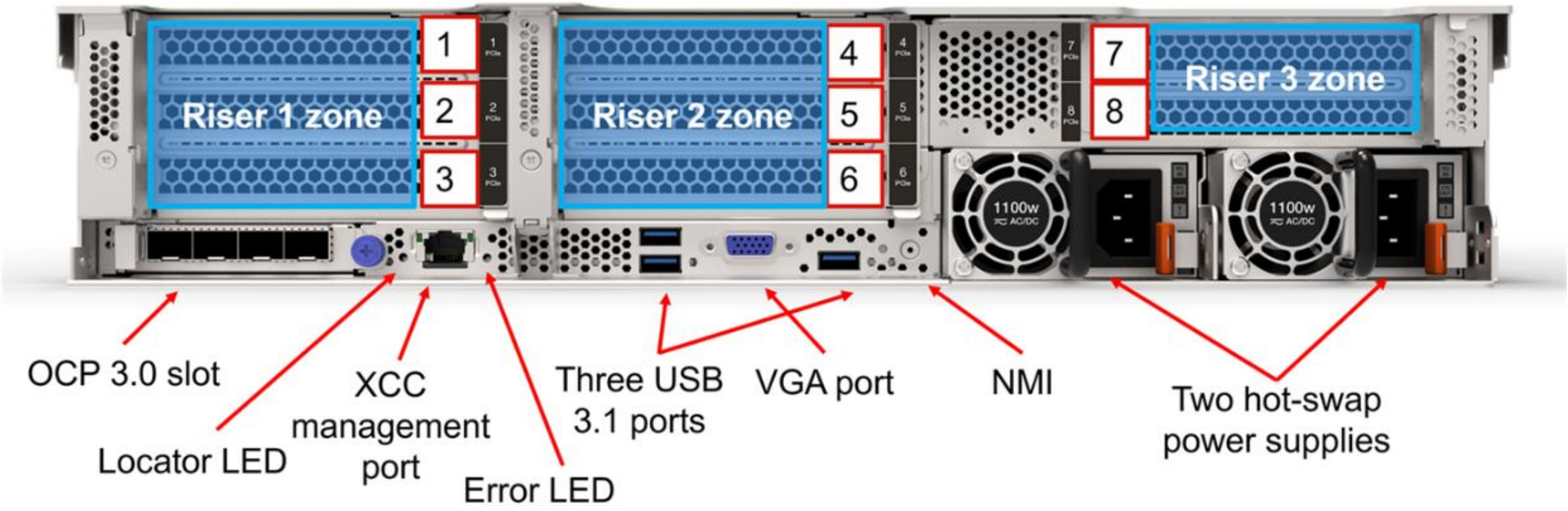
Note: For the latest specifications, refer to the Lenovo Press product guide.

ThinkSystem SR650 V3 front view

This figure shows the configuration with 24 2.5-inch drives. The other front drive configurations will be shown in the System configurations and diagrams section.



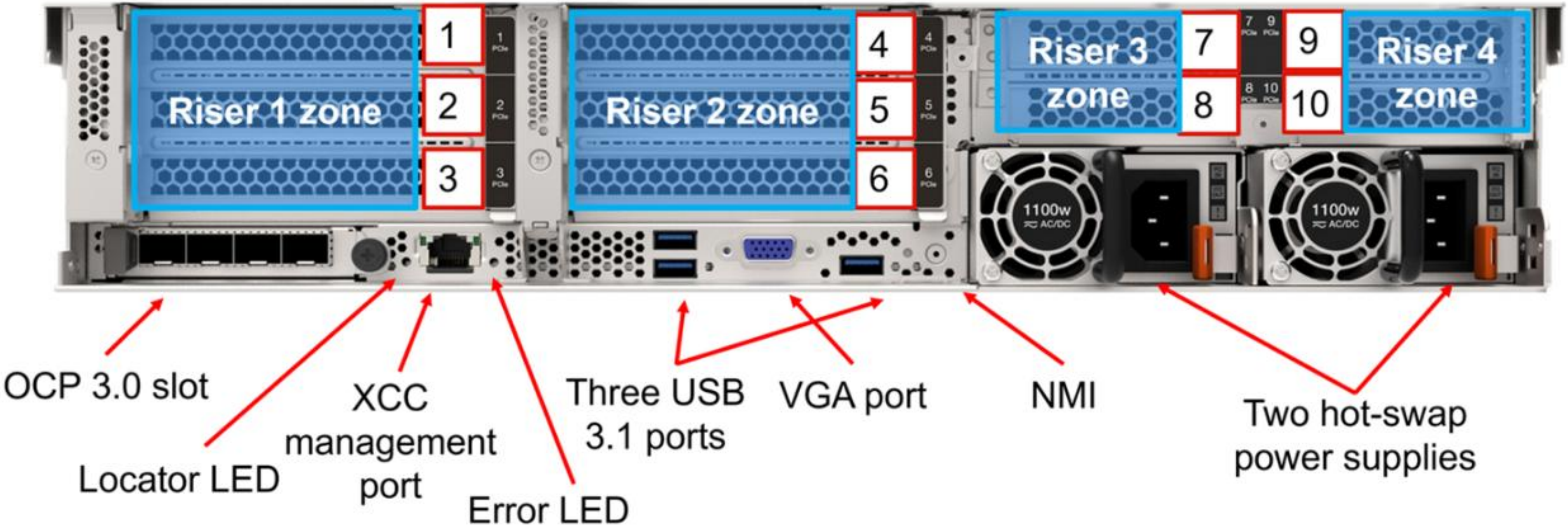
ThinkSystem SR650 V3 rear view – eight PCIe slots



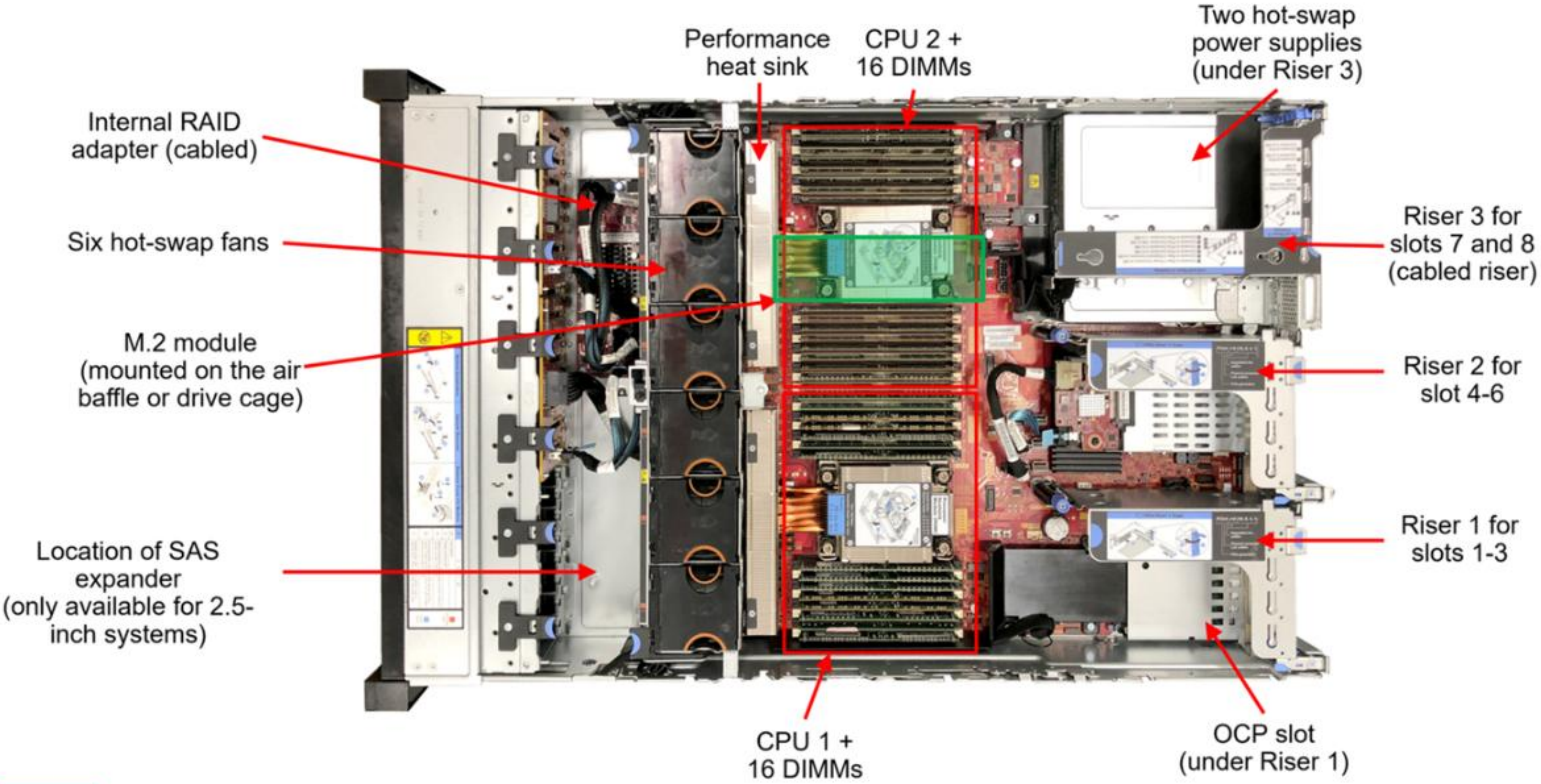
Note: For additional rear hot-swap drive and PCIe slot configuration information, refer to the Rear drive bay configurations page.

ThinkSystem SR650 V3 rear view – 10 PCIe slots

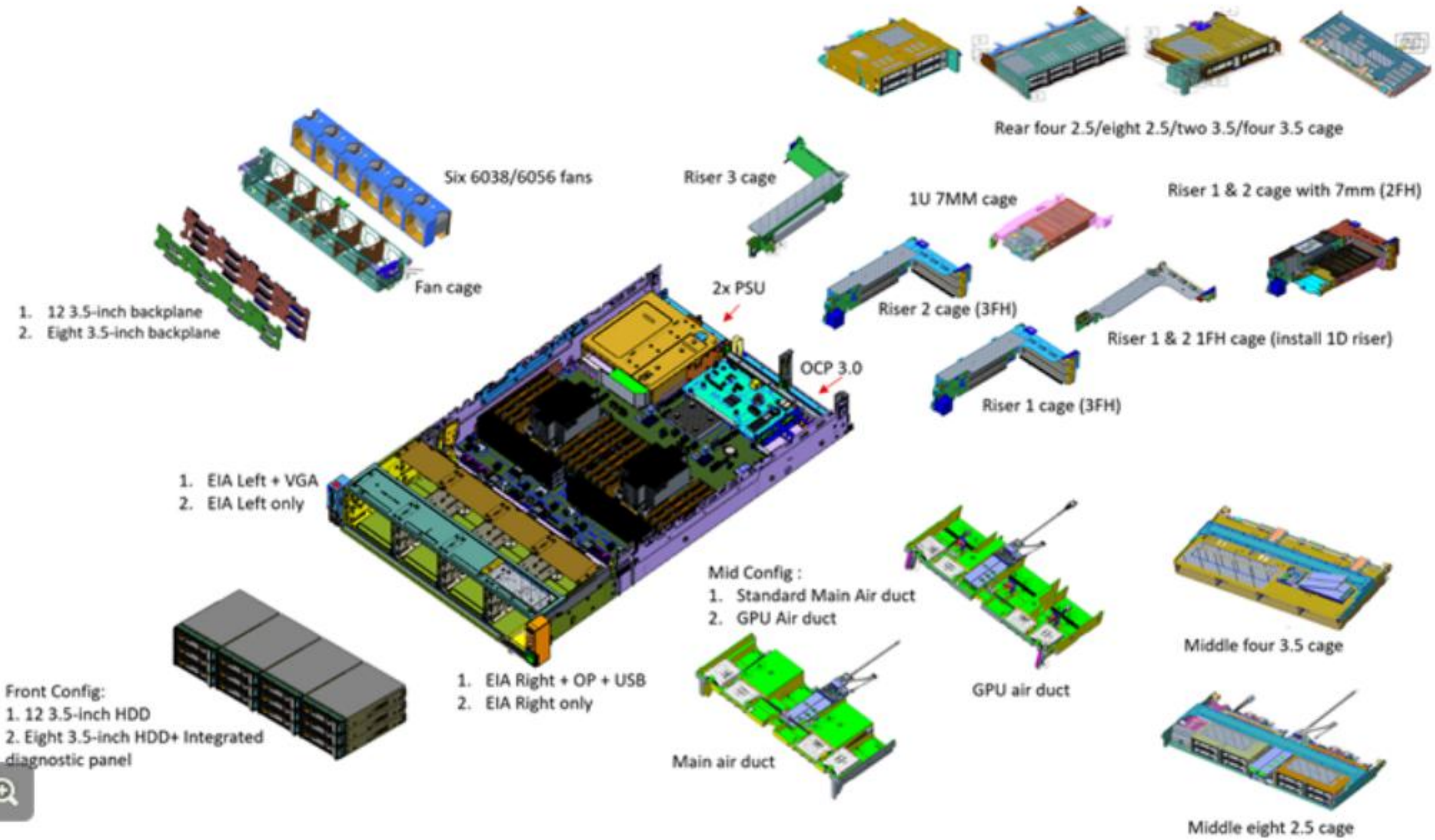
This figure shows the configuration with 10 PCIe slots.



ThinkSystem SR650 V3 inside view



ThinkSystem SR650 V3 exploded view - 3.5-inch system



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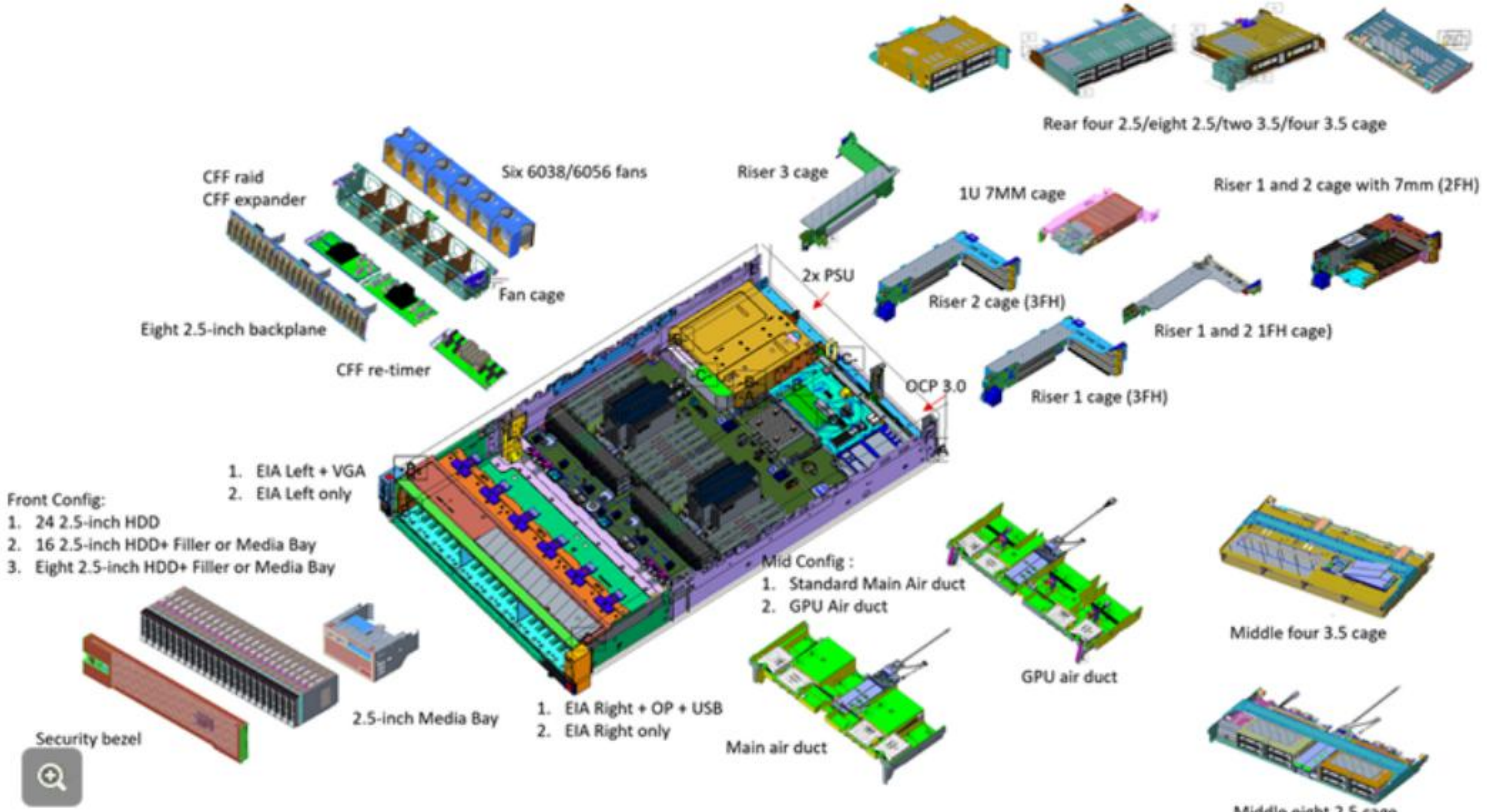
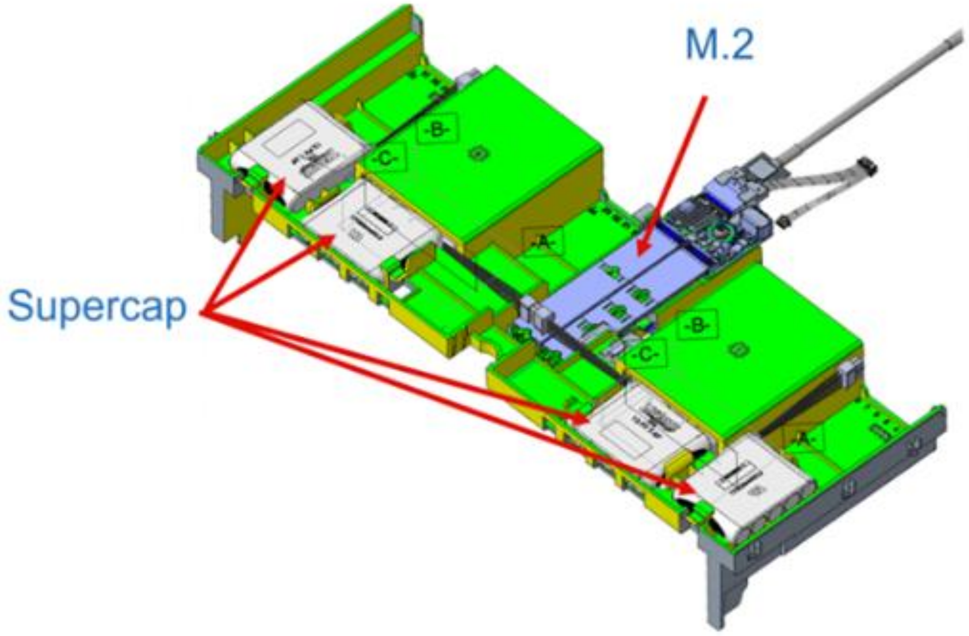
ThinkSystem SR650 V3 exploded view – 2.5-inch system

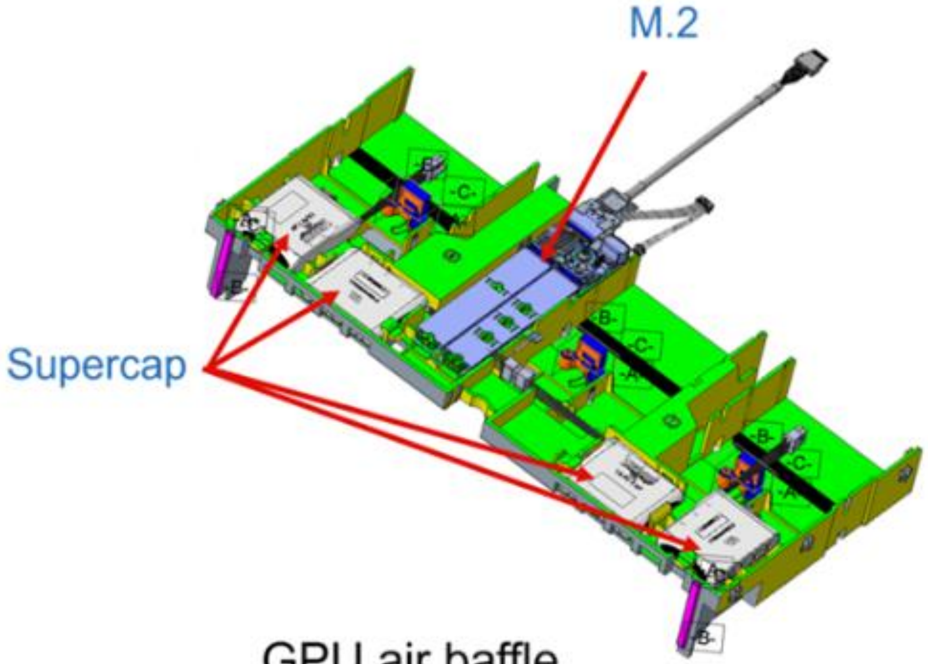


ThinkSystem SR650 V3 air baffle

The SR650 V3 supports two different air baffles. The main air baffle is the base model and should be used when GPUs or a middle drive bay are not part of the configuration. Use the GPU air baffle if they are.

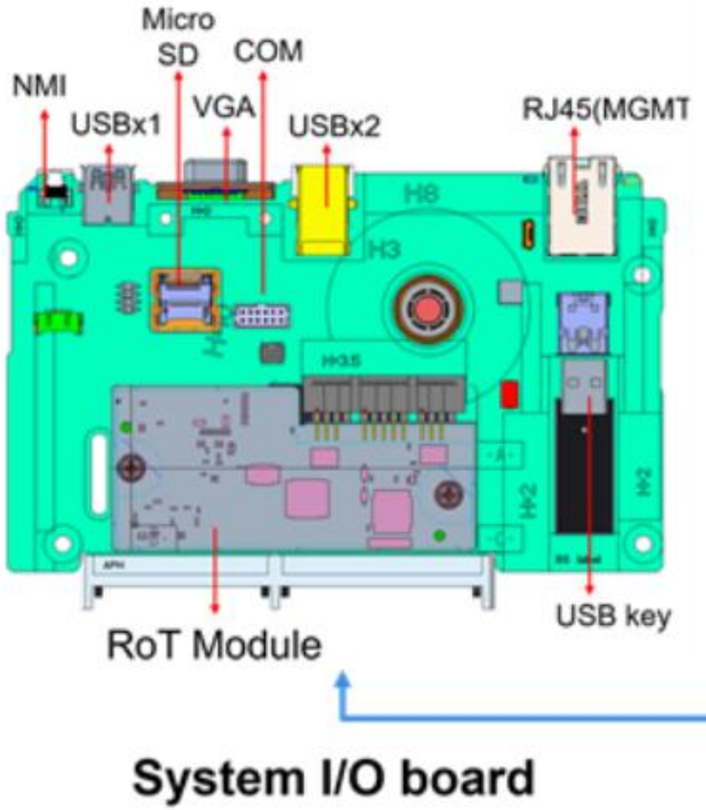


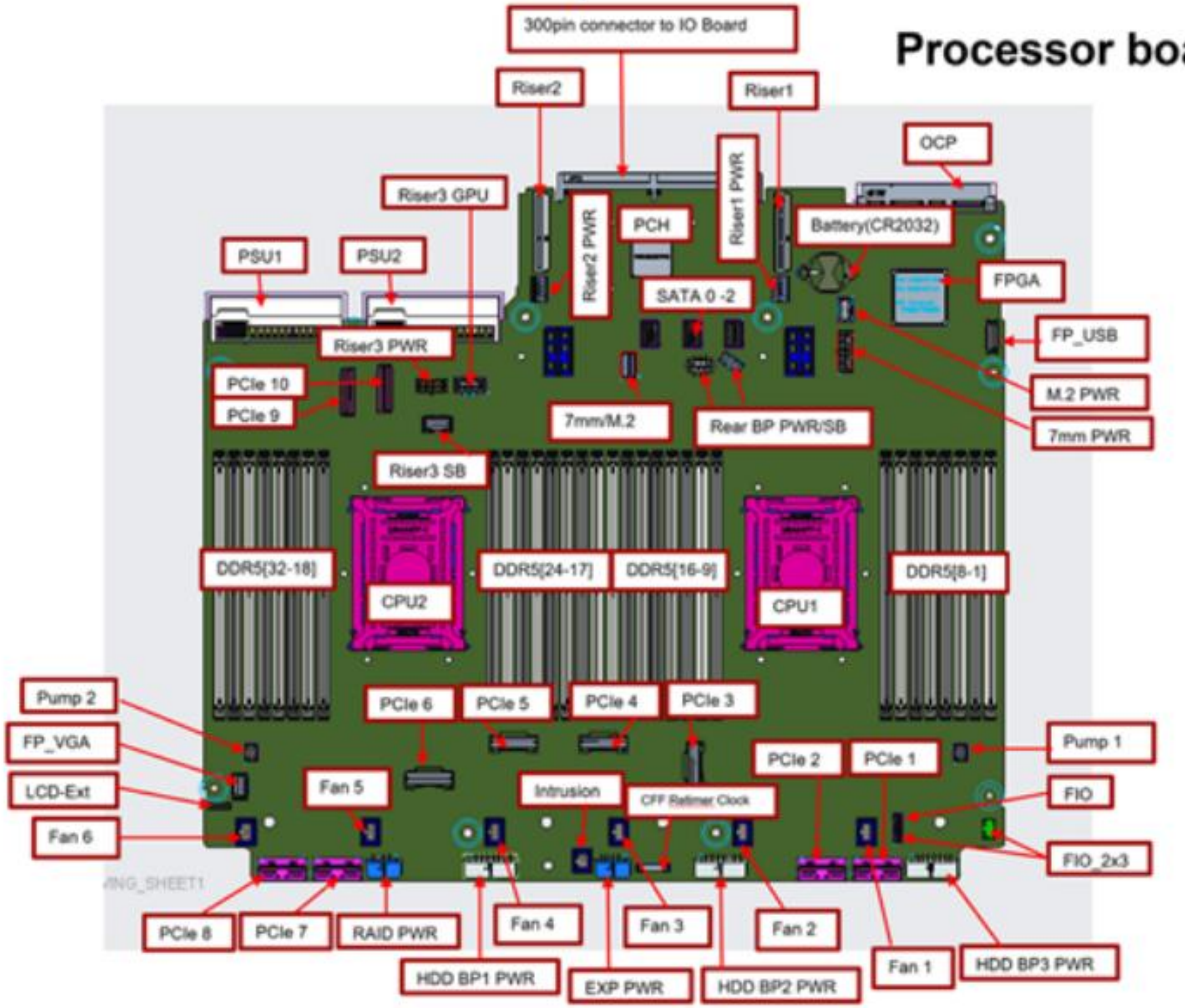
Main air baffle



GPU air baffle

ThinkSystem SR650 V3 processor and I/O board connectors





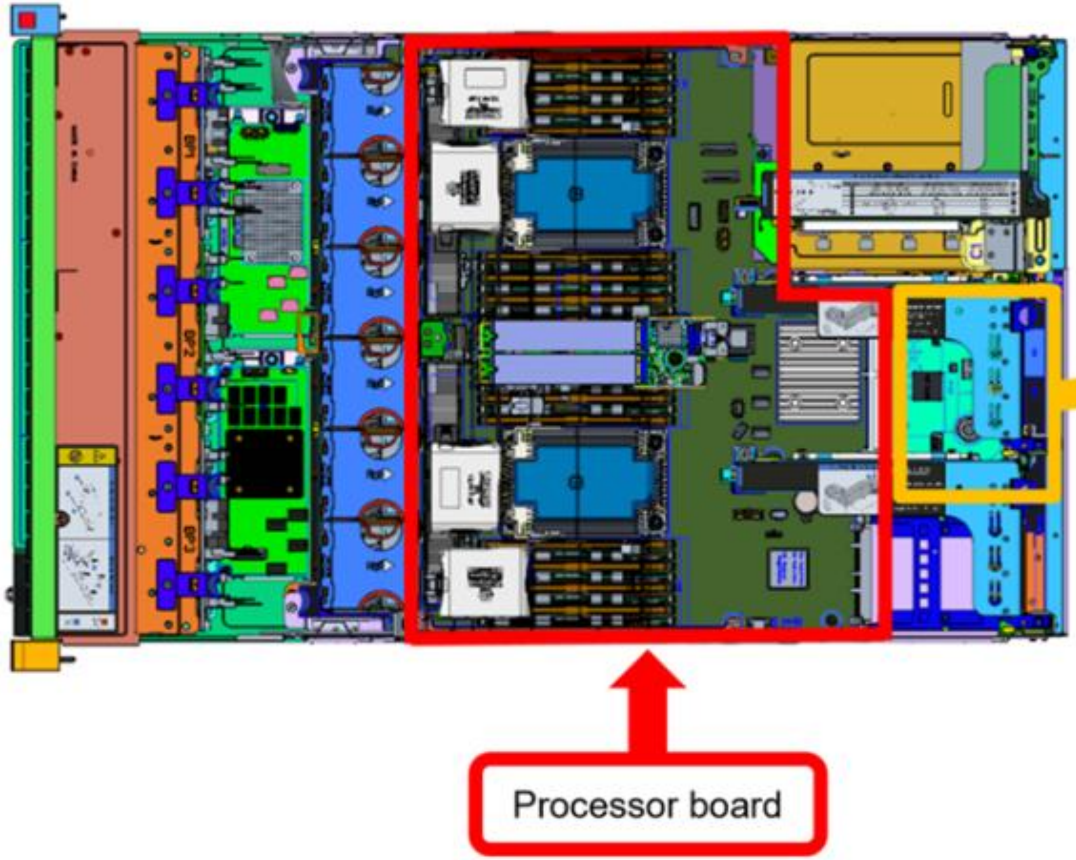
BMC I/O board and RoT module

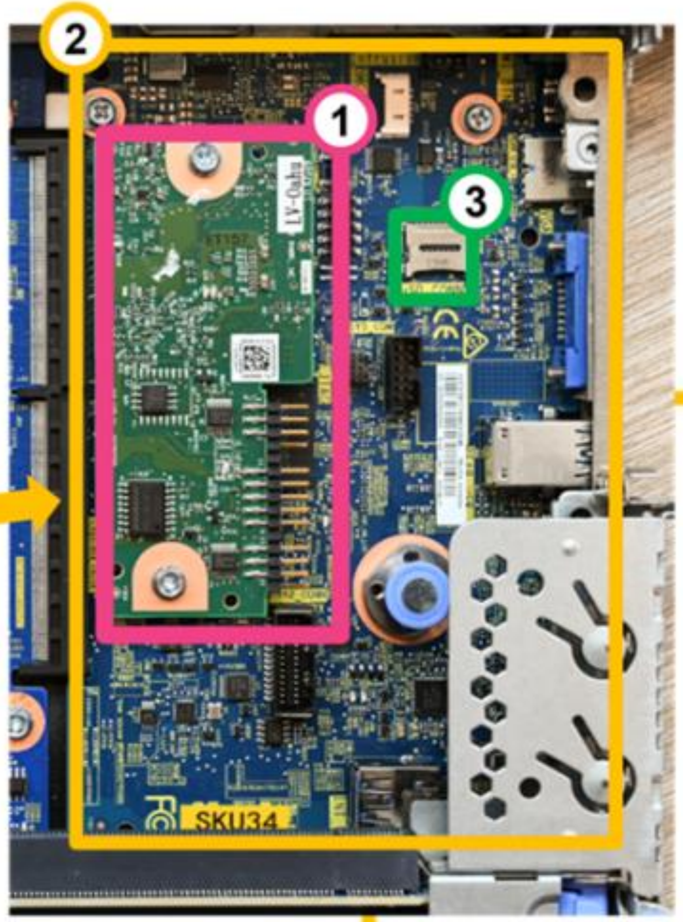
The SR650 V3 system board has three components

- Processor board
  - A board containing CPU sockets, PCIe slots, memory slots, and other server component connectors
- System I/O board
  - A board containing the system BMC (XCC2) management port, USB ports, and a VGA connector
  - A Micro SD card slot to extend XCC2 storage space for the backup of firmware and for remote console virtual media
- Firmware and Root of Trust security module (RoT module)
  - A mezzanine card containing the Trusted Platform Module (TPM), UEFI firmware, XCC2 firmware, and a silicon Root of Trust

Click HERE to see the processor board, BMC I/O board, and RoT module locations

Processor board, system I/O board, and RoT module





1 RoT module

2 System I/O board

3 Micro SD card slot

Right-side EIA latch options

When an optional integrated LCD diagnostic panel is installed, the system should be fitted with a standard EIA latch on the right. Otherwise, the right-side EIA latch with the fixed operator panel should be used.

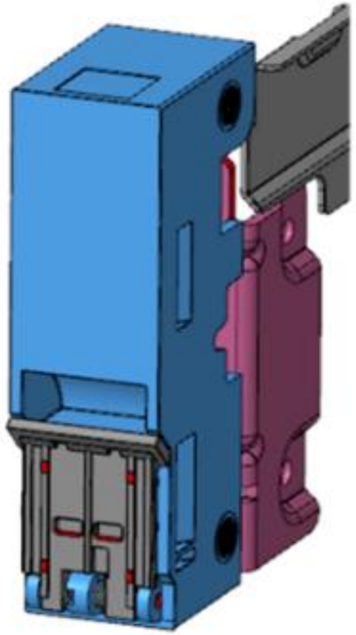
Fixed operator panel (available for all configurations)

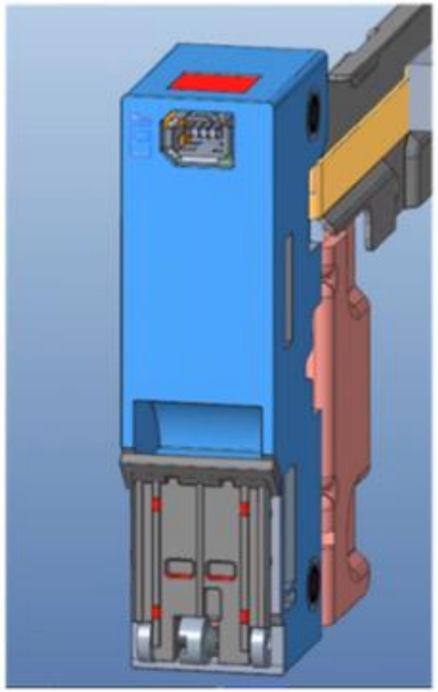
Standard EIA

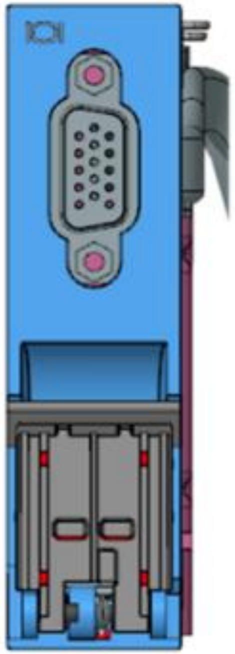
Integrated LCD diagnostic panel (eight 2.5-inch and 16 2.5-inch only)

Left-side EIA latch options

There are four options for customers to choose from:


Standard


External diagnostic port (pong)


VGA

VGA + external diagnostic port

LCD diagnostic panel

The SR650 V3 supports the external and Integrated LCD diagnostic panels. Either of the panels can be used to quickly access system information, such as active errors, system health status, firmware version, network connection status, and health information. A demo video is available on the course landing page.

External diagnostic port (pong)

Tri-Mode storage support

If installed with a RAID 940-8i or RAID 940-16i adapter, the SR650 V3 also supports NVMe through a feature called Tri-Mode support (or Trimode support). This feature enables the use of NVMe U.3 drives at the same time as SAS and SATA drives.

The following hardware components are required for Tri-Mode support:

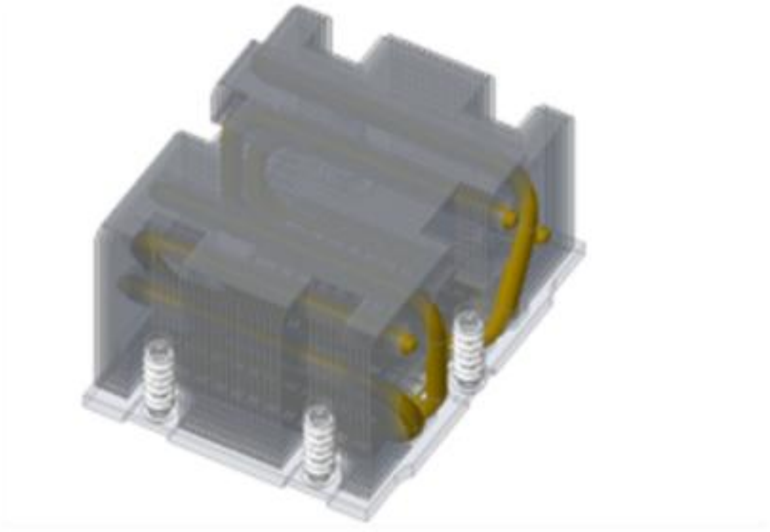
- AnyBay backplane
- RAID 940 series adapter
- U.3 drives: Only NVMe drives with a U.3 interface are supported – U.2 drives are not supported

ThinkSystem 2.5-inch U.3 NVMe PCIe 4.0 x4 HS SSD

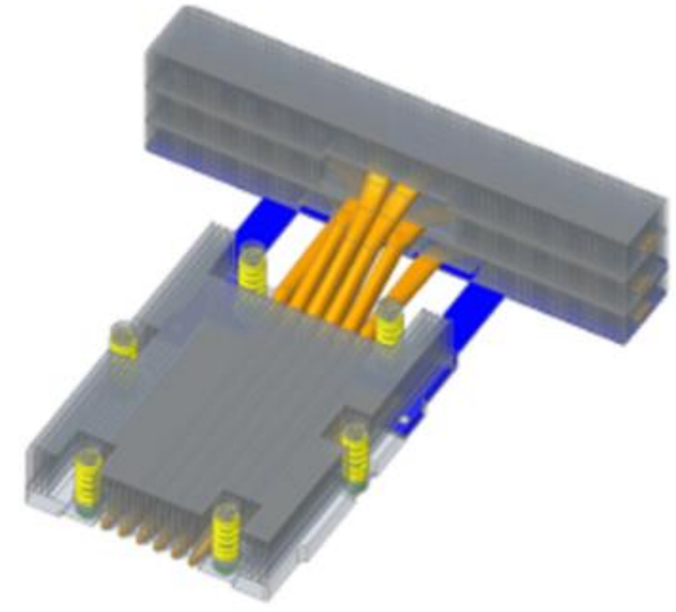
ThinkSystem RAID 940-16i Flash PCIe Gen4 12 Gb Adapter

SR650 V3 processor heat sink

The SR650 V3 supports two types of heat sink: standard and performance.



Standard 2U heat sink
For processors with a TDP of less than 300 W



Performance heat sink (1U CPU heat sink and 2U front fin)
For processors with a TDP of 300 W or more, or with middle bay drives or a GPU.